Smart Reimbursement & Local Tax Calculation Portal

Project Title:	Smart Reimbursement & Local Tax Calculation Portal
Course / Section:	Professional Electives 2 | BLOC 1
Department:	Operations / Administrative Department
Student Group:	Jerlaine Manzanero, Rosemarie Mengote, Mary Joy Lobrido, Michelle Baes, Mericar Bongbonga, Danica Abagon

1. System Title, Description, and Objectives

1.1 Workplace Context & Roles

- **Work Role:** Branch Manager / Operations Staff
- **Department / Function:** Operations & Administrative Department
- **Process Name:** Expense Reimbursement Request Process (PHP Currency with Local Tax Computation).
- **Purpose:** To verify, audit, record, and reimburse out-of-pocket expenses paid by employees using personal funds in Philippine Peso (PHP) for valid business operations, while ensuring compliance with local BIR tax requirements (12% VAT and Expanded Withholding Tax).

Users & Stakeholders

- **Requestors (Employees):** Staff members who spend personal funds in peso and submit reimbursement requests.
- **Approvers (Department Managers):** Supervisors who review and approve the business necessity and budget allocation of claims.
- **Processors & Payers (Admin / Branch Manager):** Personnel who audit attached receipts, verify computed 12% VAT and EWT amounts in peso, and disburse reimbursements.

1.2 Problem Analysis Table

#	Actual Workplace Problem	Operational Effect	Possible Cause
1	Physical Receipts Misplaced or Damaged	Delays reimbursement processing by 1 week and risks BIR audit non-compliance due to missing tax proof.	Sole reliance on paper forms and stapled physical receipts without digital backups.
2	Manual Tax Computation Errors	Inaccurate tax values lead to rejection, re-submission, and extra audit work for Finance.	Lack of automated tax computation tools and varying employee tax knowledge.

3	Lack of Real-Time Status Tracking	Employees constantly visit or message Area Managers/branch manager for updates, disrupting daily tasks.	Absence of a centralized digital portal or tracking database.
4	Dispersed Paper Records & Slow Tax Reporting	Audit spends significant manual labor compiling monthly expense and BIR tax summary reports.	Fragmented storage without a searchable, centralized database.

1.3 Proposed System & Problem-to-Solution Analysis

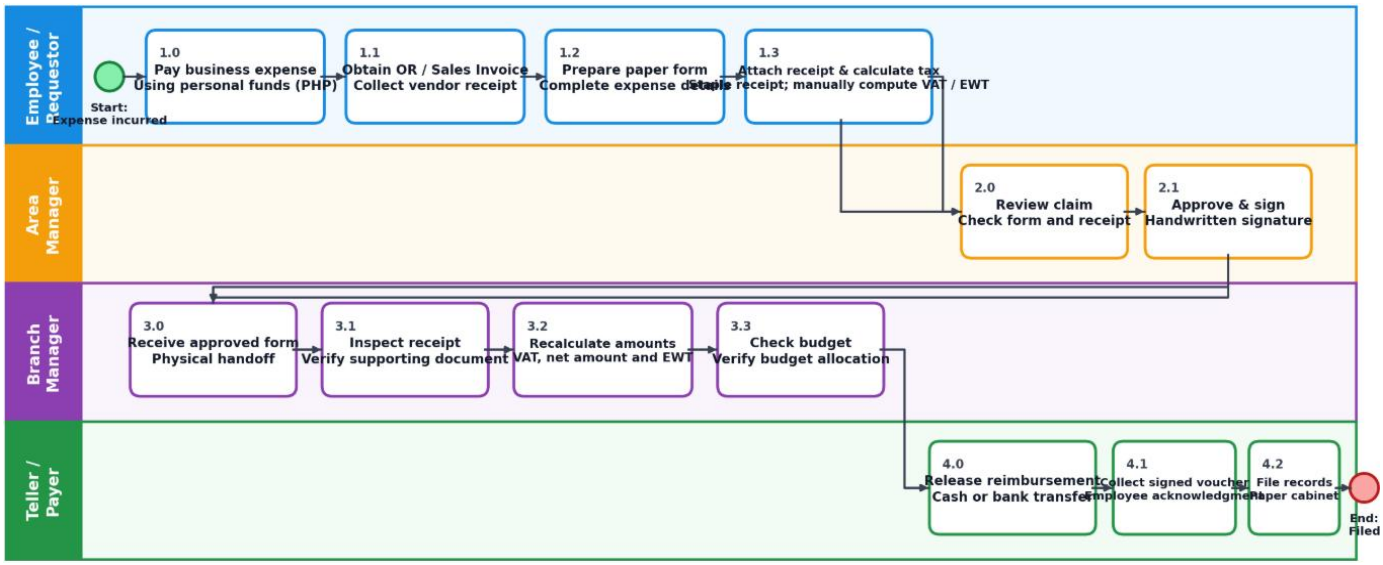
System Name: Smart Reimbursement & Local Tax Calculation Portal
System Description: A web-based application designed to digitize and automate the expense reimbursement workflow in Philippine Peso with built-in Philippine tax calculation logic.

Current Workplace Problem	Operational Effect	Proposed System Improvement
1. Manual Paper Form Submission & Stapled Physical Receipts	Receipts regularly get lost, detached, or damaged during manual routing, causing processing delays of 1 week and risking BIR audit non-compliance.	Online Submission Form with Digital Receipt Attachment: Enables staff to submit claims in PHP and upload photo/scan attachments of Official Receipts (OR).
2. Manual Local Tax Calculations (12% VAT and EWT)	Frequent manual calculation errors lead to incorrect claim totals, requiring re-submissions and extra audit rework for Finance staff.	Automated Local Tax Calculation Engine: Automatically breaks down Gross Amount, Net Amount, 12% VAT, and Expanded Withholding Tax (EWT) in peso.
3. Lack of Real-Time Status Tracking	Employees must repeatedly visit or message Managers and Finance staff to inquire about claim status, consuming productive work hours.	Real-Time Status Tracking Portal: Provides live status badges (Pending Manager Approval, Pending Payment, Paid/Disbursed) accessible at any time.
4. Dispersed Paper Records in Cabinet Folders	Historical claims are difficult to retrieve, making monthly expense audits and BIR tax report generation labor-intensive.	Centralized Digital Record Database: Stores all expense data centrally, enabling instant search, filtering, and automated tax report generation.

2. Business Process & BPMN 2.0 Modeling

2.1 As-Is Workflow (Chronological Steps)

- **Pool / Lanes:** Employee (Requestor), Department Manager (Approver), Admin Officer (Processor).
- **Start Event:** Employee pays for a business expense using personal funds in Philippine Peso.



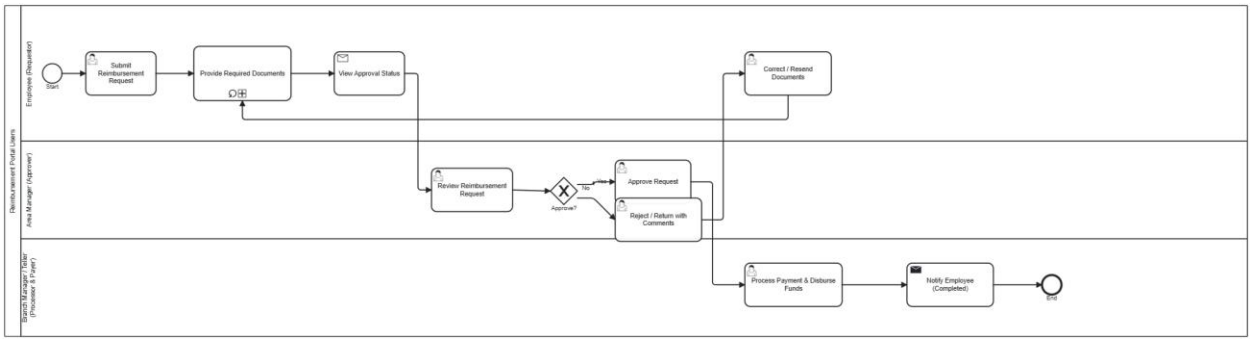
- **1. Expense Incurrence & OR Collection:** The employee pays for a business expense using personal funds in Philippine Peso (PHP) and requests an Official Receipt (OR) or Sales Invoice from the vendor.
- **2. Paper Form Filling & Manual Tax Calculation:** The employee prints a paper Expense Reimbursement Form, fills in expense details, staples the physical receipt, and manually calculates 12% VAT or EWT using a calculator.
- **3. Physical Submission & Manager Approval:** The employee physically walks the paper form and stapled receipt to their Area Manager's desk for review and signature.
- **4. Inter-Department Physical Handoff:** Once signed, the form is physically delivered to the Branch Manager, who reviews the paper form, inspects the receipt, recalculates taxes in peso, and checks budget allocations.
- **5. Payout Disbursement & Record Filing:** The teller issues cash or transfer, collects a signed disbursement voucher from the employee, and files the paper document in a storage cabinet.

2.2 To-Be Process & Operational Improvement Comparison

This model represents how the workflow operates when the **Smart Reimbursement & Local Tax Calculation Portal** is deployed, showing automated system tasks, digital receipt uploads, dynamic tax logic, and real-time tracking.

BPMN 2.0 Structural Elements (To-Be)

- **Pool 1 (User Roles / Lanes):** Reimbursement Portal Users
    - Employee (Requestor)
    - Area Manager (Approver)
    - Branch Manager / Teller (Processor & Payer)



As-Is (Current Process)	To-Be (Proposed Process)	Task Type	Direct Operational Benefit
Manual paper submission with stapled physical receipts.	Digital form submission with online receipt photo attachments.	User Task (Supported by System)	Prevents lost/damaged receipts and enables remote submission.
Manual tax calculation of 12% VAT and EWT using handheld calculators.	Automated local tax engine computing Net, 12% VAT, EWT, and Total in PHP.	Automated System Task	Eliminates calculation errors, re-submissions, and audit rework.
Physical walking of paper forms across Area Manager and Branch Manager desks.	Automated electronic routing to Manager dashboards.	Automated System Task	Eliminates transit time between desks and reduces idle delay.
Employees physically visit desks or chat to ask for	Real-Time status tracking page (Pending Approval, Pending	Automated System Task	Full visibility; eliminates constant follow-up

status updates.	Payment, Paid).		interruptions.
Manual filing of paper vouchers in filing cabinets.	Centralized database storage with automated BIR tax log reporting.	Automated System Task	Instant retrieval, audit readiness, and quick BIR tax report generation.

3. Data Flow Diagrams (DFD)

3.1 Context Diagram (Level-0 System Boundary)

The Context Diagram represents Process 0.0 (Smart Reimbursement & Local Tax Calculation System) and its external entities:

- **Employee (Requestor):** Inputs claim details (PHP), selects tax rules, uploads receipt images. Receives tracking reference IDs, status updates, and tax breakdowns.
- **Area Manager (Approver):** Receives pending claims list and receipt previews. Inputs approval/rejection decisions with review notes.
- **Branch Manager / Teller (Processor & Payer):** Receives verified approved queue. Inputs payout confirmation (Cash/Bank Transfer). Receives monthly BIR tax audit reports.

3.2 Level-0 DFD Decomposition & Data Stores

- **D1: Reimbursements —** Stores request details, monetary amounts in PHP, calculated tax breakdowns (Net, 12% VAT, EWT), receipt image URLs, and current status.
- **D2: Approvals & Audit Logs —** Stores approval history, manager comments, decision timestamps, and teller payout logs.

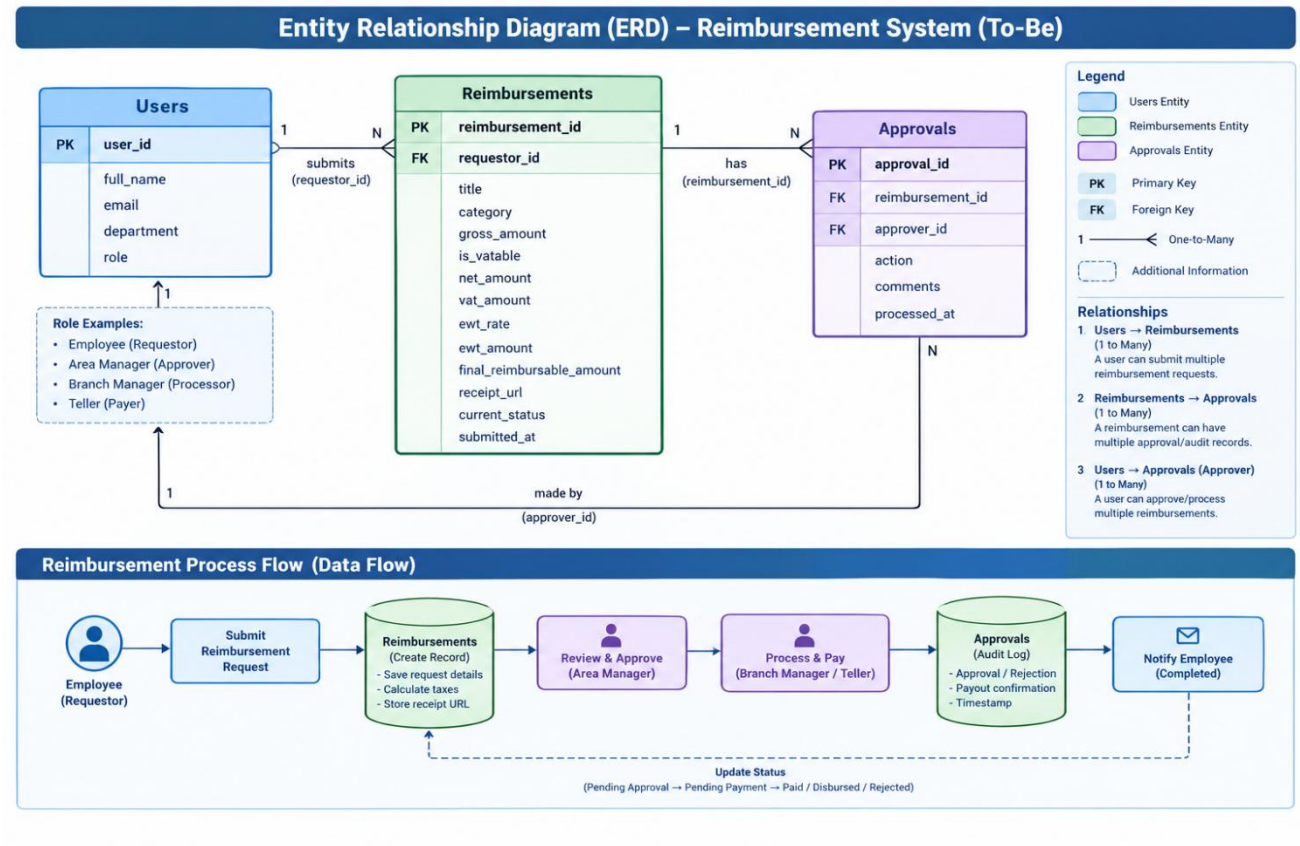
Process Details & Tax Computation Logic

- **Process 1.0 (Submit Claim & Compute Tax):** Receives claim title, gross amount in PHP, receipt file, and tax toggles. Computes net, 12% VAT, EWT, and final reimbursable amount. Writes to D1.
- **Process 2.0 (Review & Approve Claim):** Reads pending claims from D1. Captures Area Manager decision & comments. Writes audit log to D2 and updates status in D1.
- **Process 3.0 (Disburse Payout & Log Tax):** Retrieves claims pending payment from D1. Verifies computed tax amounts and processes payout. Writes log to D2 and updates D1 status to Paid.
- **Process 4.0 (Track Status & Generate Reports):** Queries D1 and D2 to display real-time tracking badges to employees and monthly BIR tax summary reports to managers/tellers.

Philippine Tax Calculation Engine Formulas (VAT-Inclusive)

- Net Amount = Gross Amount / 1.12 (If VATable)
- 12% VAT Amount = Gross Amount - Net Amount
- EWT Amount = Net Amount * EWT Rate (Rates: 0%, 1%, 2%, 5%)
- Final Reimbursable Amount = Gross Amount - EWT Amount

4. Entity Relationship Diagram (ERD)



Entity Name	Primary Key / Foreign Keys	Attributes & Description
Users	PK: user_id	full_name, email, department, role (Employee, Area Manager, Branch Manager, Teller). Stores user authentication & roles.

Reimbursements	PK: reimbursement_id FK: requestor_id -> Users.user_id	title, category, gross_amount, is_vatable, net_amount, vat_amount, ewt_rate, ewt_amount, final_reimbursable_amount, receipt_url, current_status, submitted_at.
Approvals	PK: approval_id FK: reimbursement_id -> Reimbursements FK: approver_id -> Users.user_id	action (Approved/Rejected), comments, processed_at. Associative entity providing a complete 1:M audit trail.

4.1 System Artifact Consistency Check Matrix

Design Artifact	Question to Verify	Verification & System Mapping
To-Be BPMN	Which activities must users accomplish through the application?	1. Requestors submit PHP claim details, select tax rules, upload receipt images. 2. Area Managers review pending claims, inspect digital receipts, submit approval/rejection with notes. 3. Tellers audit tax values, disburse funds in PHP, and mark claims Paid.
Context / Level-0 DFD	What information is received, processed, stored, and sent?	Inputs: Claim details, PHP Gross Amount, Tax Flags, Receipt Image, Decisions, Payout Status. Processed: Tax Engine logic (Net, VAT, EWT, Final Amount), status transitions. Stored: Saved in D1 (Reimbursements) and D2 (Approvals). Outputs: Live status badges, pending queues, BIR tax reports.
ERD	Where will each required record and relationship be stored?	Users entity stores profiles & roles. Reimbursements entity stores claim attributes and persistent tax computations. Approvals entity logs decision history and timestamps.

5. System Features and Functionalities

- **Role-Based Interface & Access Control:** Persistent header switcher allowing seamless role previewing between Employee, Area Manager, and Branch Manager/Teller.
- **Interactive Dashboard Metrics:** Real-time cards displaying Total Submitted, Pending Approvals, Total Disbursed, Total 12% VAT, and Total EWT in PHP (₱).

- **Automated Local Tax Engine:** Form with live recalculation of Net Amount, 12% VAT, EWT, and Final Reimbursable Amount upon Gross input or tax flag toggles.
- **Digital Receipt Preservation:** Secure photo/scan receipt attachment stored as URL previewable inside detail modals.
- **Live Status Tracking:** Real-time color-coded badges: Pending Approval (Orange), Pending Payment (Blue), Paid (Green), Rejected (Red).
- **Automated SMS Alerts:** Automated SMS notification integration (e.g., Semaphore / Twilio) sending alerts to Area Managers on submission and Employees on approval/disbursement.
- **BIR Tax Report Generator:** Monthly aggregation view of VAT and EWT amounts from paid claims for Philippine tax filing compliance.

6. Synthetic Test Cases & Execution Evidence

Test Case	Expected Result	Actual Result / Execution Evidence	Pass/Fail
Create Transaction	Valid record is saved and tax breakdown is computed accurately.	Submitted claim for Office Supplies (Gross: ₱500.00) by Jerlaine on Oct 29, 2026. System calculated Net: ₱446.43, 12% VAT: ₱53.57, EWT (0%): ₱0.00, Final Reimbursable: ₱446.43. Saved under Ref REB-2026-KV9V.	PASS
Retrieve Transaction	Saved record is retrieved with full details and receipt preview.	Opened modal for Ref REB-2026-KV9V. System correctly retrieved all stored tax parameters, user profile, category, tax breakdown values (₱500.00 Gross / ₱446.43 Net / ₱53.57 VAT), and receipt image preview.	PASS
Update / Process Action	Change persists and status badge updates dynamically.	Area Manager processed transaction decision by setting status badge to Rejected. Status badge dynamically turned red, payout fields remained unexecuted (—), and state persisted accurately across reloads.	PASS

7. Operational Impact & Conclusion

> **Key Workplace Improvements Achieved**
>
> 1. Complete Elimination of Reimbursement Delays & Desk Walk-Overs: Shifted workflow from 1-2 weeks paper routing to instant online submission and routing.
> 2. 100% Tax Computation Accuracy (12% VAT & EWT): Automated tax logic eliminates handheld calculator

errors and guarantees BIR audit readiness.

3. Total Real-Time Visibility & Full Auditability: Live status badges eliminate constant status inquiry interruptions and maintain permanent approval logs.

Screenshot of the Developed System

Apps URL: https://literate-clear-claim-flow.base44.app

